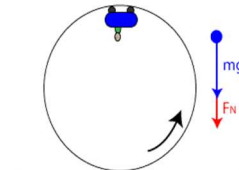
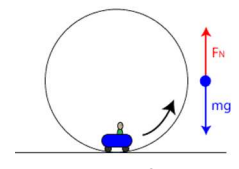


Top of Circle



$$N = \frac{mv^2}{r} - mg$$
$$\text{Critical Speed (N=0)} \\ v = \sqrt{rg}$$

Bottom of circle



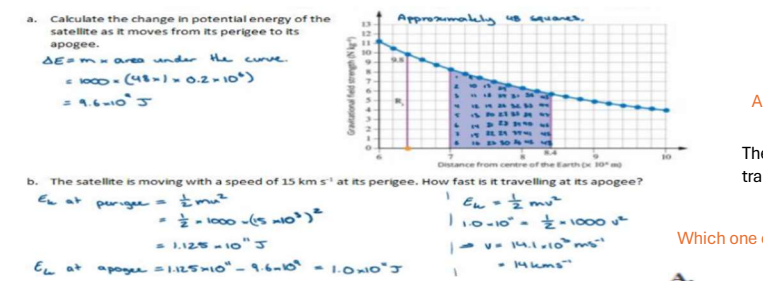
$$N = mg + \frac{mv^2}{r}$$

Projectile Motion

$$\text{Max Height: } S_v = \frac{(u \sin \theta)^2}{2g}$$
$$\text{Height: } S_v = s_h \tan \theta + \frac{1}{2}at^2$$
$$\text{Range: } S_h = \frac{u^2 \sin 2\theta}{g}$$
$$\text{Time of flight: } t = \frac{2u \sin \theta}{g}$$

→ If the projectile beings its path from the max height (e.g, a cliff) don't multiply by 2

A decommissioned satellite of mass 1000 kg has an elliptical orbit around the Earth. At its closest approach (its perigee), it is 600 km above the Earth's surface. At its furthest point (its apogee) it is 2000 km from the Earth's surface. The Earth has a radius of 6.4×10^6 m



Match each force to an arrow on the diagram for the direction of the force when the car is faster and slower. Label the following forces: Gravity force on the car, Normal reaction force of the road on the car, The sideways frictional force on the tyres from road.

Faster	Slower
B - F_{friction}	D - F_{friction}
C - F_g	C - F_g
A - F_N	A - F_N

Example Question 5

A stone is thrown from the top of a 15 m high cliff above the sea at an angle of 30° to the horizontal. It has an initial speed of 20 ms^{-1} . Ignore air resistance.

Calculate the time taken for the stone to reach the sea.

$$t = \frac{u \sin \theta}{g} = \frac{20 \sin 30}{10} = 1$$
$$s = ut + \frac{1}{2}at^2 \rightarrow 20 = 0 + 5t^2 \rightarrow t = 2$$

Total time = 1 + 2 = 3 seconds

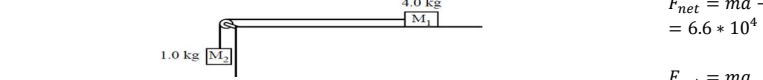
$$s_v = \frac{(u \sin \theta)^2}{2g} = \frac{(20 \sin 30)^2}{20} = 5$$

Calculate the magnitude and direction of the velocity of the stone immediately before it reaches the sea. Give the direction as the magnitude of the angle between the velocity and the horizontal.

$$v_h = 20 \cos 30 = 17.32 \text{ ms}^{-1}$$
$$v = u + at = 0 + 10(2) = 20 \text{ ms}^{-1}$$
$$20^2 + 17.32^2 = v^2 \rightarrow v^2 = 700$$
$$\rightarrow v = 26.5 \text{ ms}^{-1}$$
$$\tan \theta = \frac{20}{17.32} \rightarrow \theta = 49.1^\circ$$
$$26.5 \text{ ms}^{-1} \text{ at an angle of } 49.1^\circ$$

Example Question 7

Students set up an experiment as shown below. M_1 , of mass 4.0 kg, is connected by a light string to a hanging mass, M_2 , of 1.0 kg. The system is initially at rest. Ignore the mass of the string and friction. The masses are released from rest.



- Calculate the acceleration of M_1
 - Calculate the magnitude of the tension in the string as the masses accelerate.
- $$m_2 g - T = m_2 a$$
- $$1 \times 10 - T = m_2 a$$
- $$T = 4a = 4 \times 2 = 8 \text{ N}$$

Example Question 9

Two blocks, A of mass 4.0 kg and B of mass 1.0 kg, are being pushed to the right on a smooth, frictionless surface by a 40 N force.

Calculate the magnitude of the force on block B by block A (F on B by A).

$$F_{\text{net}} = ma \rightarrow 40(4.0 + 1.0)a \rightarrow a = 8 \text{ ms}^{-1}$$

To accelerate block B at this rate you need 8N of force.

State the magnitude and the direction of the force on block A by block B (F on A by B).

Force must be in the opposite direction. Therefore, 8N Left.

Theory

Newton's First Law: An object's velocity is constant unless a net external force acts on it. If $F_{\text{net}} = 0$, then acceleration is zero.

Newton's Second Law: A net force causes an object to accelerate in the direction of the force ($F_{\text{net}} = ma$).

Newton's Third Law: For every action, there is an equal and opposite reaction. Action-reaction pairs act on different objects, are of the same type, and are equal in magnitude.

Apparent Weight (F_N): The contact force that creates the sensation of weight.

Accelerating up: $F_N > F_g$ (feel heavier).

Accelerating down: $F_N < F_g$ (feel lighter).

Weightlessness (Freefall): The sensation when gravity is the only force acting ($F_N = 0$). You are accelerating downwards at g. The gravitational force (F_g) still exists.

Terminal Velocity: Reached when the upward drag force equals the downward gravitational force. $F_{\text{net}} = 0$, so acceleration is zero and velocity is constant.

In an isolated system (no external net forces), total momentum is conserved ($\Sigma p_{\text{initial}} = \Sigma p_{\text{final}}$). Momentum lost from an object (e.g., due to friction) is transferred to the Earth.

Reducing Impact Force: Safety features (airbags, crumple zones) increase the impact time (Δt). For a given change in momentum (Δp), this decreases the average net force (F_{net}), reducing injury.

The net work done on an object equals its change in kinetic energy ($W_{\text{net}} = \Delta E_K$).

Elastic Collisions: Momentum and kinetic energy are conserved.

Inelastic Collisions: Momentum is conserved, but kinetic energy is not (converted to heat, sound).

Energy on an Incline: Down a frictionless incline: GPE $\rightarrow$ KE. With friction: GPE $\rightarrow$ KE + Heat.

Effect of Air Resistance: A drag force that opposes motion. It reduces the maximum height and range, creating an asymmetrical trajectory (steeper descent).

Requirement: A net force (the centripetal force) must act towards the centre of the circle.

Centripetal Force (F_c): The resultant force, not a fundamental force. It can be provided by friction, tension, gravity, or a component of the normal force.

Velocity is tangential. Acceleration and net force are central. Because the force is always perpendicular to the velocity, no work is done, and speed is constant.

Vertical Loops:

Top (upside down): $F_c = F_g + F_N$. You feel lighter (FN is small). Minimum speed to stay on the track is when $FN=0$.

Bottom: $F_c = F_N - F_g$. You feel heavier (FN is large).

An object can only continue vertical circular motion if $N \geq 0, v \geq v_c$

Banked Turns: The horizontal component of the normal force provides the centripetal force. An ideal bank requires no friction.

Conical Pendulum: If the speed is doubled this would increase the tension in the string and the horizontal net force. Force of gravity remains constant on the ball. Hence the angle with the horizontal would decrease. However, the radius of rotation would also increase.

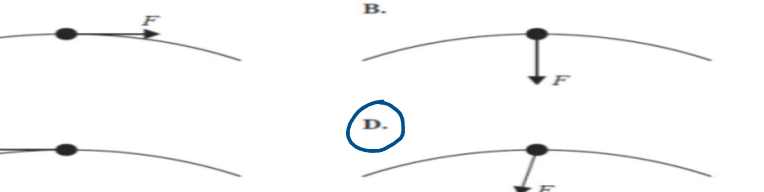
Example Question 2

A Melbourne tram goes around a corner. The track follows a circular path. The force that supplies the tram with the necessary centripetal acceleration is:

The curved rails and wheel flanges provide an inward reaction force, the centripetal force, that directs the tram around the corner. The motor maintains speed, but the rails change direction.

Example Question 4

Which one of the following diagrams best shows the direction of the resultant force, F, on the ball at the position of maximum height in the real situation where air resistance is not ignored?



Example Question 6

A tennis ball is hit horizontally from a height of 1.8 m above the ground with an initial speed of 24 m/s. The ball travels towards a net that is 12 m away and 0.90 m high.

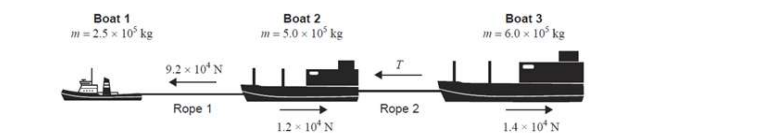
Assuming there is no air resistance, calculate how far above the net the ball will be when it passes over it. Show your working.

$$t = \frac{12}{24} = 0.5 \text{ ms}^{-1}$$
$$s = ut + \frac{1}{2}at^2 = 0 + 4.9 \times 0.5^2 = 1.225 \text{ m}$$

Starts from a height of 3.6m, so it will be 3.6-1.225=2.375m above the ground. It will clear the net by 1.475 m.

Example Question 8

Boat 1, a tug boat, with a mass of 2.5×10^5 kg, is towing two larger boats. Boat 2 has a mass of 5×10^5 kg and Boat 3 has a mass of 6×10^5 kg. A resistance force of 1.4×10^5 N acts on Boat 2 and a resistance force of 1.4×10^4 N acts on Boat 3. The tension in Rope 1 is 9.2×10^4 N.



Show that the acceleration of the two boats being towed, Boat 2 and Boat 3, is $6.0 \times 10^{-2} \text{ ms}^{-2}$

$$F_{\text{net}} = ma \rightarrow F_{\text{net}} = 9.2 \times 10^4 - (1.2 \times 10^4 + 1.4 \times 10^4)$$
$$6.6 \times 10^4 = (5.0 \times 10^5 + 6.0 \times 10^5)a$$
$$a = 0.06 \text{ ms}^{-2} = 6.0 \times 10^{-2} \text{ ms}^{-2}$$

Calculate the tension, T, in Rope 2. Show your working.

$$F_{\text{net}} = ma$$
$$T_{\text{rope2}} - \text{Friction} = 6.0 \times 10^5 \times 6.0 \times 10^{-2}$$
$$T_{\text{rope2}} = 6.0 \times 10^5 \times 6.0 \times 10^{-2} + 1.4 \times 10^4$$
$$T_{\text{rope2}} = 5.0 \times 10^4 \text{ N}$$

Example Question 10

A golf ball near the edge of a cliff is hit at a speed of 48 ms^{-1} and at an angle of 35° to the horizontal. The golf ball lands on the ground 6.2 s after it is hit. Ignore air resistance.

Determine the height, h, of the cliff.

$$h = ut + \frac{1}{2}at^2 = 48 \sin 35 \times 6.2 + (-4.9) \times 6.2^2 = -17.85$$
$$\therefore h = 18 \text{ m}$$

Calculate the horizontal distance, x, that the golf ball travels before it reaches the ground below.

$$s = vt = 48 \cos 35 \times 6.2 = 240 \text{ m}$$

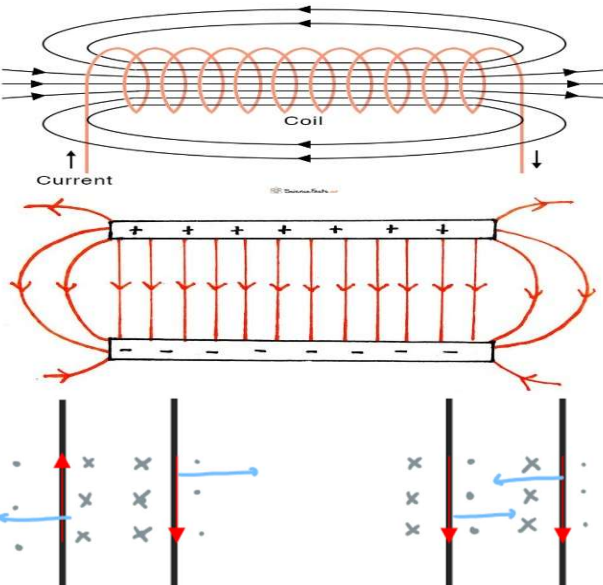
Describe how, in real life, air resistance would affect the golf ball's flight path.

In real life, air resistance will take energy out of the system. Therefore, the ball will not travel as high or as far.

Formulae

$$G \frac{M * m}{r^2} = \frac{mv^2}{r}$$
$$t = \sqrt{\frac{4\pi^2 r^3}{GM_{\text{sun}}}}$$
$$a = \frac{4\pi^2 r}{T^2}$$
$$\frac{1}{2}mv^2 = G \frac{M_{\text{sun}}m}{2r}$$
$$v = \sqrt{G \frac{M_{\text{sun}}}{r}} = \frac{2\pi r}{t}$$
$$r = \sqrt[3]{\frac{GM_{\text{sun}}t^2}{4\pi^2}}$$
$$\tau = r_{\perp} F$$
$$E = vB$$
$$\frac{r_1^3}{t_1^2} = \frac{r_2^3}{t_2^2}$$

Solenoid Magnetic Field



Example Question 1

Find the expression for the altitude above Earth's surface (radius R) where the gravitational field strength is half its surface value.

$$= (\sqrt{2}R) - R$$

Example Question 2

A hollow satellite with all its mass, M, located at a radius R from its centre. What graph best represents the force of gravity experienced by an astronaut at a distance r ($0 < r < \infty$) from the centre of the satellite?



Example Question 4

Explain why geostationary satellites must be vertically above the equator to remain stationary relative to Earth's surface.

All orbits must be centred on Earth's centre of mass due to gravity. The only orbital path centred on Earth's core that also perfectly matches the Earth's surface rotation without any north-south drift is an orbit directly above the equator. Any other inclination would cause the satellite to move north or south relative to the ground observer.

Example Question 6

What would happen if the split ring in a DC motor was replaced with a slip ring. Instead of continuous rotation in one direction, the motor would:

Rotate approximately half a turn (180 degrees) due to the initial torque.

Then, experience a reversing torque that opposes the rotation.

The motor would likely oscillate back and forth around the vertical position where the torque changes direction. It might rock a bit past the vertical due to inertia, but the reversed torque would quickly push it back.

Example Question 8

2 large isolated parallel metal plates are placed at distance, d, apart. What happens to the magnitude of the electric field midway between the plates if d is halved?

The electric field increase by a factor of 2.

What happens to the magnitude of the potential difference between the two plates if d is doubled.

The voltage has not changed.

Example Question 10

Calculate the magnitude and direction of the electric field that would be required for an electron to travel through undeflected. (at a velocity of 1.1×10^{-7} m/s and magnetic field strength of 1×10^{-4} T)

$$E = vB = (1.1 \times 10^7) \times (1 \times 10^4) = 10^{11} \text{ NC}^{-1}$$

Example Question 11

Explain how the ICON satellite maintains a stable circular orbit without the use of propulsion engines.

Once placed in orbit, the satellite will continue to fall to earth under the influence of the Earth's gravitational field. The gravitational field is at right angles to the satellites motion and exerts a constant force radially inwards on the satellite. Therefore, the satellite will move in a circular path. No engines are required to keep it in orbit (free fall).

Theory

Field lines go from N to S, + to -. (For gravity, the mass always attracts) Density of the field lines indicates strength (closer = stronger).

Field lines CANNOT touch.

Use the right-hand grip rule to determine the direction of the magnetic field. Use the right-hand slap rule to determine the force acting on the conductor.

All forces are vectors, and calculations should account for this.

In a stable circular orbit, momentum changes but kinetic energy remains constant.

If mass is halved, the work will also be halved because $w = fd$ and force is proportional to the mass.

For a satellite to be geostationary, it must; have a period of 24 hours, orbit directly above the equator, and orbit same direction as earth's rotation (W to E).

In line with Kepler's Third Law (insert law), a satellite's period is unaffected by changes in its own mass because the period depends only on the mass of the central celestial body it orbits and the distance from that body's centre of mass.

The inverse square law states that the intensity a physical quantity radiating from a point source (like gravity) decreases proportionally to the inverse of the square of the distance from that source ($g \propto \frac{1}{r^2}$)

Gravitational field strength is 0 N kg^{-1} at the centre of Earth due to appeal to symmetry as the vector sum of all the mass of the Earth would be zero.

To find ΔGPE , calculate area under the graph.

When calculating the area under the graph, be mindful of the units. (F-d: don't multiply by mass, GFS-d: multiply by mass) $\text{Joules} = \text{N} \cdot \text{m}$

Electric field and force are constant in parallel plates and change in distance between the plates doesn't affect voltage.

DC motors convert electrical energy into rotational kinetic energy (rotational motion)

If a charge moves in the direction, it would naturally go (e.g., a positive charge moving towards a negative source), the electric field does work on the charge.

If a charge is forced to move against its natural direction, work must be done on the electric field.

If current flows parallel in the same direction as the magnetic field, the force is zero.

Parallel wires - Current flowing in the same direction attracts while opposite directions repel.

A split-ring commutator's function is to reverse the direction of the current in the coil every half rotation. This reversal ensures the torque continues to act in the same rotational direction, allowing for continuous spinning instead of just oscillating back and forth.

In particle accelerators, electric field control the kinetic energy and speed, whereas the magnetic field controls the direction, as it acts perpendicular to the velocity.

The magnitude of torque experienced by the coil is at its maximum value when the coil is parallel with the magnetic field.

The magnitude of torque is zero when; the plane of the coil is perpendicular to the direction of the magnetic field.

The split-ring commutator reverses the direction of current flow within the rotating coil every half turn, precisely when the coil passes the vertical position, it ensures that the forces acting on the sides of the coil always produce a torque that drives the motor continuously in the same direction. Without it, the torque would reverse every half-turn, preventing continuous rotation.

A solenoid acts like a bar magnet, with a distinct North and South pole. Field lines on the outside act much like the field lines on a bar magnet. With field lines on the inside being straight and parallel.

The field strength of a solenoid can be increased by current or turn density.

Dots are assumed as towards you (i.e. out of the page) & crosses as away from you (i.e. into the page).

According to the right-hand slap rule moving charged particles within a magnetic field will always move in a circular arc as the magnetic force acting on them is constant & perpendicular to the direction of motion.

Your palm would indicate the force direction for a positive charge. Since the electron is negative, the force (FB) is in the exact opposite direction to where your palm faces. The proton will curve in the direction your palm is pushing.

Example Question 3

Singly charged positive ions, each with a mass of 2.18×10^{-26} kg and a charge of 1.60×10^{-19} C, are accelerated from rest through a potential difference of 1500 V. They then enter a uniform magnetic field of 0.10 T, which is perpendicular to their direction of motion, causing them to follow a semicircular path.

Calculate the diameter of this path.

$$v = \sqrt{\frac{2qV}{m}} = 3.16 \times 10^5$$
$$r = \frac{mv}{qB} = \frac{(4.80 \times 10^{-27}) \times (3.16 \times 10^5)}{(1.6 \times 10^{-19}) \times 0.10} = 9.4 \times 10^{-2} \text{ m}$$

Example Question 5

Two point charges, Q and 4Q, are placed 12 cm apart, as shown in the diagram below. On the straight line between the charges Q and 4Q, the electric field is



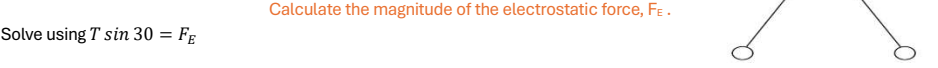
Zero at a point 4 cm from Q.

$$\frac{kQ}{r^2} = \frac{k4Q}{(12-r)^2} \Rightarrow r = 4$$

Example Question 7

Two equally charged balls, each of mass 4.0 g, from light, non-conducting strings suspended from a low ceiling. The charged balls repel each other with the strings at an angle of 60° , as shown below.

Show that the tension force, T, in each string is 4.5×10^{-2} N.



Calculate the magnitude of the electrostatic force, F_E .

Example Question 9

A stationary electron is in a uniform magnetic field between two parallel plates. A switch is closed, creating an electric field that accelerates the electron. Ravi says there will be a magnetic force of constant magnitude but changing direction. Mia says there will be a constantly increasing magnetic force in the same direction. Evaluate these two statements.

Mia is correct about the increasing magnitude of the force, because the electric force acting on the charge will cause it to accelerate. This in turn means that the magnetic force will increase in size as the electron speeds up, using $F = Bqv$

Ravi is correct in saying that the direction of the force will be continually changing. Using the right hand slap rule, the magnetic force will always be perpendicular to the direction of the motion, therefore, the direction of this force will also change.

Therefore, Mia is correct about the size of the force, but incorrect about its direction, and Ravi is correct about the direction of the force, but incorrect about its magnitude.

Example Question 12

An electron velocity selector consists of an electric field, E, perpendicular to a magnetic field, B. Electrons at the required velocity, v_0 , pass straight through, while others are deflected.

Show that the velocity of an undeflected electron is given by $v_0 = E/B$

$$F = Eq \text{ and } F = Bqv \rightarrow Eq = Bqv \rightarrow v = E/B$$

At which point (X, Y, or Z) would electrons travelling faster than v_0 arrive? Explain your answer.

Z. The force due to the magnetic field varies with v. Therefore, the magnetic force down the page will increase as the speed increases. The force due to the electric field remains constant. Therefore, the force down is greater than the force upwards.

